

THz Biosensor Based on Metasurface Integrated with Spintronic Emitter

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Abstract—Currently spintronic THz sources have the advantages of low cost, high efficiency, and easy integration, so it can be used to integrate feature-rich devices. In this paper, we demonstrate the function of THz sensing by metasurface integrated with spintronic THz emitter. Through electromagnetic simulation, the resonant frequency point of metasurface was designed to be around 0.8 THz. Then the metasurface was fabricated on a quartz substrate and shared the substrate with a spintronic emitter. This integrated THz biosensor was used to measure the transmittance spectra of water, alcohol and olive oil. The experimental results show that it is sensitive for identifying different liquids according to the frequency shift.

I. INTRODUCTION

Terahertz band is between 0.1 THz and 10 THz. It is a special region between electronics and photonics, and it is a transition field from macroscopic to microscopic. Terahertz wave has the characteristics of low energy, good penetration, good coherence and so on. Terahertz has great application value and academic research value because of its unique frequency characteristics.

With the emergence of artificial electromagnetic materials and the rapid development of metasurface artificial microstructure, it is possible to design highly sensitive terahertz sensors, which also promotes the application of terahertz technology in biomedical field. Metasurface can be used to detect biomolecules, proteins and various solutions [1, 2]. Spintronics is a subject that studies the transport properties of electrons by controlling their spin and constructs novel devices. Some physical phenomena of spintronics, such as exchanging-type magnetovibrators,

antiferromagnetic resonance, and ultrafast spin mechanics, have characteristic frequencies in the terahertz band. The combination of terahertz and spintronics forms a new interdisciplinary subject called terahertz spintronics, which promotes the development of the two disciplines. Using the corresponding spintronics phenomena and principles, researchers have discovered and established a new kind of terahertz source, named spintronic THz emitter, has been developed with a very simple structure [3]. This kind of THz source can be prepared by magnetron sputtering or molecular beam epitaxy method, which has the advantages of low cost and high efficiency in THz generation. A new type of THz biosensor can be formed by fabricating spintronic THz source and sensor structure on the front and rear surfaces of a dielectric substrate respectively.

In this paper, the structure of THz biosensor is schematically illustrated in Fig. 1. The metasurface was processed on one side of the substrate, and the Fe/Pt nanofilms were processed on the other side of the substrate. The sample was placed in a static in-plane magnetic field. The portable THz time-domain spectroscopy system was built for the experimental test. The experiment uses a photoconductive antenna to detect terahertz signal. The THz wave carrying information was emitted from source and transmitted through the metasurface, then detected by the THz detector, as shown in Fig. 2. we used femtosecond linearly polarized laser pulses with a pulse width of 100 fs, a central wavelength of 780 nm and pumping power of 52mW.

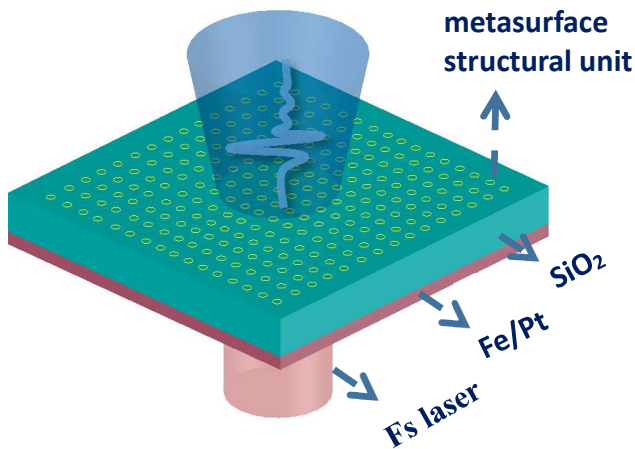


Fig. 1. Femtosecond laser pulses incident onto the Fe/Pt/SiO₂/metasurface structure for THz generation and sensing.

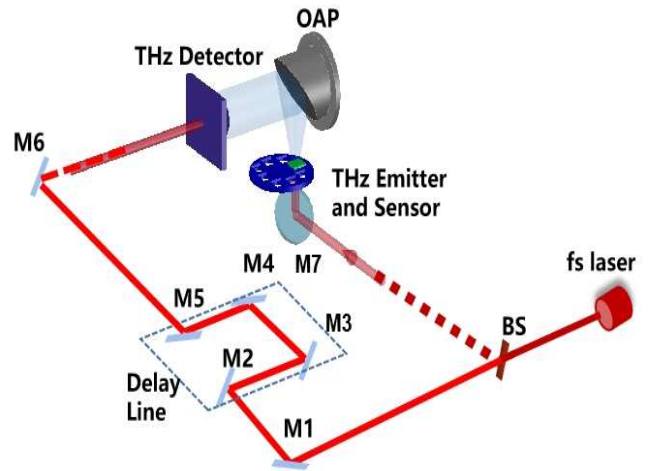


Fig. 2 THz time-domain spectroscopy system.

II. RESULTS

The experimental results are shown in Fig.3. the time domain diagram of the metasurface are shown in Fig. 3(a). As shown in Fig. 3(b), the transmittance spectra of air, water, alcohol and olive oil were measured with the fabricated biosensor respectively. We can see that the different solutions have different resonant frequency point. According to the simulation results and the experimental data, the change of the dielectric properties of measured liquids is the main reason of different resonant frequency points. The sensor sensitivity is expressed as the frequency shift of the resonant peak per unit refractive index. Water, alcohol and olive oil have different refractive indices and dielectric constants, so they can be easily identified using this sensor. This demonstrates that such kind of biosensors can also be used to detect biological or chemical samples for the further research.

III. CONCLUSION

In this paper, a sensor composed of ferromagnetic nanofilm and microstructured metasurface was designed. Three kinds of liquid samples with different refractive indices were measured by this sensor. The experimental results show that their resonance absorption peaks are different, so we can identify the liquid samples well. These results provide a method for the applications of terahertz sensors in the fields of biological and chemistry for label-free detection.

ACKNOWLEDGMENTS

This work was supported in part by NSAF under grant No. U1930127 and Fundamental Research Funds for the Central Universities under grant No. ZYGX2019J013.

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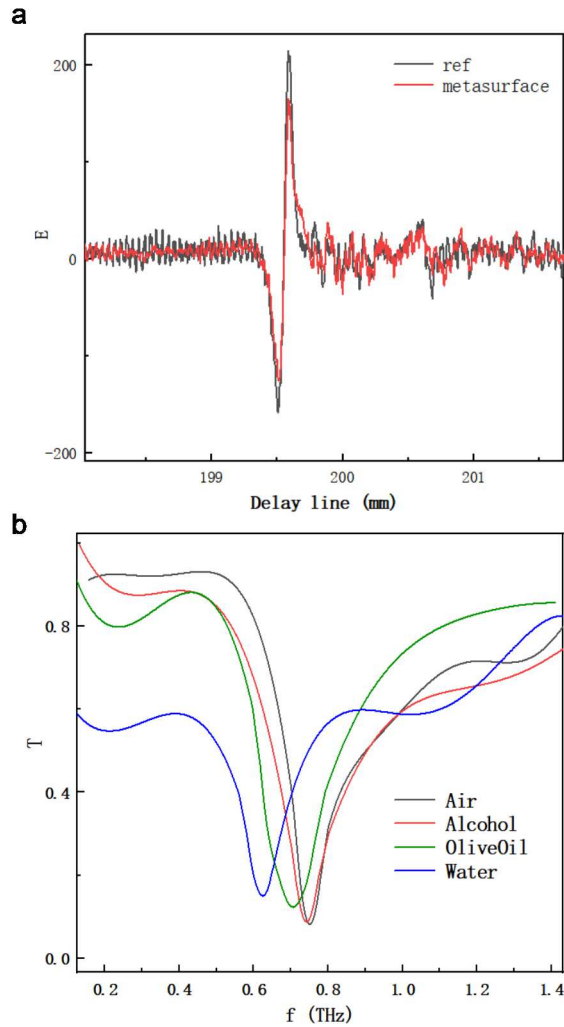


Fig.3. (a) Time domain signals of air reference and metasurface. (b) Experimental results of sensor tests for air, water, alcohol and olive oil.